

CPP_REFERENCE_INDEX

Core Definition

Index for the C++ standard library quick-reference set. Each file follows the same shape: a **Target | Operation | Output** signature table, common patterns with runnable examples, important concepts, and best practices.

#CPP #REFERENCE #INDEX #STDLIB

FILES

Requested set

File	Header	Covers
IOSTREAM_REFERENCE.md	<iostream>	Stream hierarchy, cin / cout / cerr / clog, formatted and unformatted I/O, stream state, format flags, tie/rdbuf, sentries
IOMANIP_REFERENCE.md	<iomanip> + <ios>	Every manipulator — parameterized and not. setw / setfill / setprecision, alignment, base, float format, quoted, put_time / put_money, stickiness rules, format-state RAI
FSTREAM_REFERENCE.md	<fstream>	ifstream / ofstream / fstream / filebuf, open modes and the mode matrix, positioning, binary I/O, random access, filesystem companions
CSTRING_REFERENCE.md	<cstring>	C-string and raw-memory functions, wide counterparts, the safety cheat sheet, why std::string usually wins
STRING_REFERENCE.md	<string>	std::string full member list, string_view, searching, numeric conversion, SSO, iterator invalidation, performance table
CMATH_REFERENCE.md	<cmath>	All math functions, classification, rounding, special functions (C++17), <numbers> constants, floating-point comparison, domain table

Standard I/O library (per en.cppreference.com/w/cpp/io)

File	Header	Covers
IOS_REFERENCE.md	<ios>	ios_base and basic_ios — the foundation. All flag types, state semantics, copyfmt, xalloc / iword / pword, callbacks, io_errc
SSTREAM_REFERENCE.md	<sstream>	istringstream / ostringstream / stringstream / stringbuf, parsing and building, C++20 view() and move-str(), tool-selection table
STREAMBUF_REFERENCE.md	<streambuf>	The buffer model, get/put areas, every protected virtual, working custom buffers (null sink, tee, prefixer, buffered in/out), char_traits
CSTDIO_REFERENCE.md	<cstdio>	FILE* I/O, full printf / scanf specifier tables, binary I/O, RAI wrapper, the classic traps, mapping to std::format

File	Header	Covers
MODERN_IO_REFERENCE.md	<code><print></code> <code><format></code> <code><syncstream></code> <code><spanstream></code> <code><iosfwd></code>	C++20/23 I/O: format spec grammar, custom formatter, thread-safe output, allocation-free streams, forward declarations, removed headers

WHERE TO LOOK

Question	File
"Why does my getline read an empty line?"	IOSTREAM (the >> then getline trap)
"How do I make a column line up?"	IOMANIP (aligned table output)
"Why is my setw only working once?"	IOMANIP (stickiness table)
"What does ios::ate do vs ios::app?"	FSTREAM (open mode matrix)
"How do I read a whole file into a string?"	FSTREAM (slurp)
"Is strncpy safe?"	CSTRING (the strncpy trap)
"How do I split a string?"	STRING (splitting) / SSTREAM
"Why doesn't 0.1 + 0.2 == 0.3?"	CMATH (comparing floats)
"What's the difference between fail and bad?"	IOS (state flag decision table)
"How do I reset a stringstream?"	SSTREAM (the two-step reset)
"How do I make cout write to a socket?"	STREAMBUF (custom buffers)
"Why is my printf crashing?"	CSTDIO (printf is not type-safe)
"How do I print from multiple threads?"	MODERN_IO (osyncstream)
"How do I format my own type?"	MODERN_IO (custom formatter)
"Which header do I include in my .hpp?"	MODERN_IO (iosfwd)

STANDARD VERSION SUMMARY

C++11	move semantics on streams; string paths for fstream; getline; stoi/to_string; cbrt/hypot/round/isnan and the classification family; io_errc; scoped enums
C++14	std::quoted; ""s literals; gets removed
C++17	filesystem::path in fstream; string_view; charconv; special math functions; 3-argument hypot; clamp
C++20	std::format; syncstream; stringstream view()/move-str(); numbers constants; starts_with/ends_with; lerp; bit_cast; three-way comparison
C++23	std::print/println; spanstream; string::contains; resize_and_overwrite; ios::noreplace; range formatting
C++26	stringstream removed; println(); runtime_format

COMPILER FLAGS WORTH USING

-std=c++20	Or c++23 where the toolchain supports it
-Wall -Wextra -Wpedantic	The baseline
-Wformat=2	Catches printf/scanf mismatches
-Wshadow -Wconversion	Catches quiet correctness bugs
-fsanitize=address,undefined	Finds the buffer and UB bugs these headers invite
-D_GLIBCXX_ASSERTIONS	libstdc++ bounds checks in debug builds

EXTERNAL RESOURCES

- **Input/Output library:** <https://en.cppreference.com/w/cpp/io>

- **Strings library:** <https://en.cppreference.com/w/cpp/string>
 - **Numerics library:** <https://en.cppreference.com/w/cpp/numeric>
 - **C++ Core Guidelines:** <https://isocpp.github.io/CppCoreGuidelines/CppCoreGuidelines>
 - **Compiler support tables:** https://en.cppreference.com/w/cpp/compiler_support
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